

Jennifer Gonzales-Pasion

| jgonzal2@andrew.cmu.edu | (908) 635-6163 | www.linkedin.com/in/jennifer-gonzales-pasion

EDUCATION

Carnegie Mellon University

M.S. in Mechanical Engineering | GPA: 4.00/4.00

Pittsburgh, PA
Expected May 2027

Stevens Institute of Technology

B.Eng. in Chemical Engineering with Honors | GPA: 3.50/4.00

Hoboken, NJ
May 2025

TECHNICAL SKILLS

Programming: Python, Bash, C++, Javascript, MATLAB, Linux environments

Computational Materials: Density Functional Theory (DFT), Classical Molecular Dynamics (MD), Machine-Learning Interatomic Potentials (MLIPs)

AI & Machine Learning: Large Language Model (LLM) Agents, Dimensionality Reduction (t-SNE)

Quality & Validation: Performance Qualification, Equipment Validation, Good Manufacturing Practice (cGMP)

INDUSTRY EXPERIENCE

Colgate-Palmolive: Hill's Pet Nutrition

Richmond, IN / Topeka, KS

Quality Intern

Jun 2026 – Aug 2026

- Executed Performance Qualification (PQ) protocol on packaging metal detectors, establishing challenge diets based on fiber, fat, and density to ensure food safety compliance
- Performed consumer complaint investigation by analyzing plant cGMP records, foreign material log, and work orders
- Conducted internal audit by evaluating First-In, First-Out (FIFO) material handling procedures and inventory tracking, mitigating expiration risks and reducing scrap costs

Quality Intern

Jun 2025 – Aug 2025

- Authored and executed PQ protocol for high-speed PECO filling inspection systems, ensuring strict adherence to quality and cGMP standards
- Evaluated operational downtime and line status of critical X-ray food safety equipment to identify bottlenecks
- Presented validation findings and data-driven product loss reduction strategies to cross functional leadership

RESEARCH EXPERIENCE

Mechanical & AI Lab

Carnegie Mellon University

Research Assistant

Oct 2025 – Present

- Developing an AI agent leveraging LLMs and machine learning force fields to accelerate catalyst design and battery materials
- Evaluating the Universal Model for Atoms (UMA) potential across the ElectroFace Dataset, benchmarking performance and accuracy against ab-initio molecular dynamics (AIMD)

Hensley Catalyst Research Laboratory

Stevens Institute of Technology

Research Assistant

Dec 2023 – May 2025

- Operated USPEX code, perform density functional theory calculations using VASP and write python scripts for data collection automation.
- Generated initial training data through USPEX to support the subsequent development of a machine learned interatomic potential for Ni-Au catalysts.

Sandia National Laboratories

Albuquerque, NM

Computational Multiscale Intern

Jun – Aug 2024

- Developed and validated nonmagnetic MLIP for Ni-Au catalysts to investigate surface reconstruction and active site formation using software tools VASP, FitSNAP, Dakota, and LAMMPS
- Achieved accurate predictions of stable pure Ni and Au surfaces, contributing to the understanding of Ni-Au catalysts for hydrogen fuel cells; presented findings at 3 symposiums

HONORS AND AWARDS

- **Fellowship & Internship:** GEM Employer Fellow | DOE Student Undergraduate Laboratory Intern (2024)
- **Stevens Institute of Technology Awards:** Edwin A. Stevens Scholarship, Martha Bayard Stevens Scholarship, Torrance Stevens Scholarship, and Presidential Scholarship
- **Academic Awards:** Dean's List (2023 Spring – 2025 Spring)